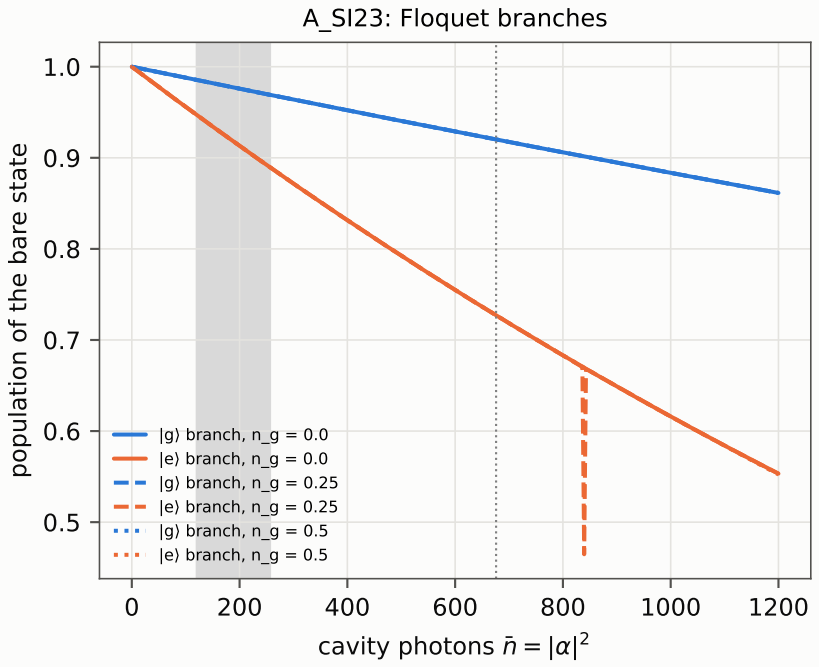
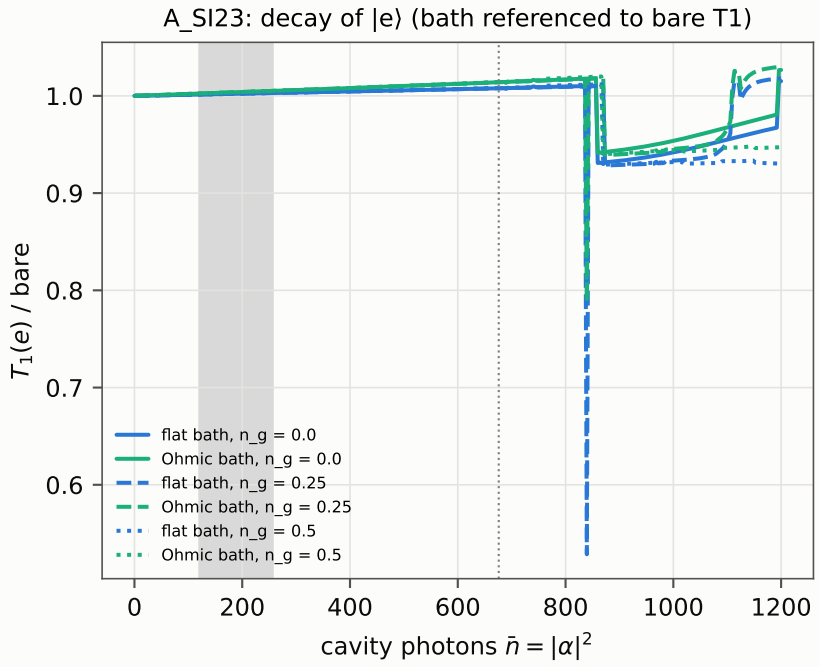
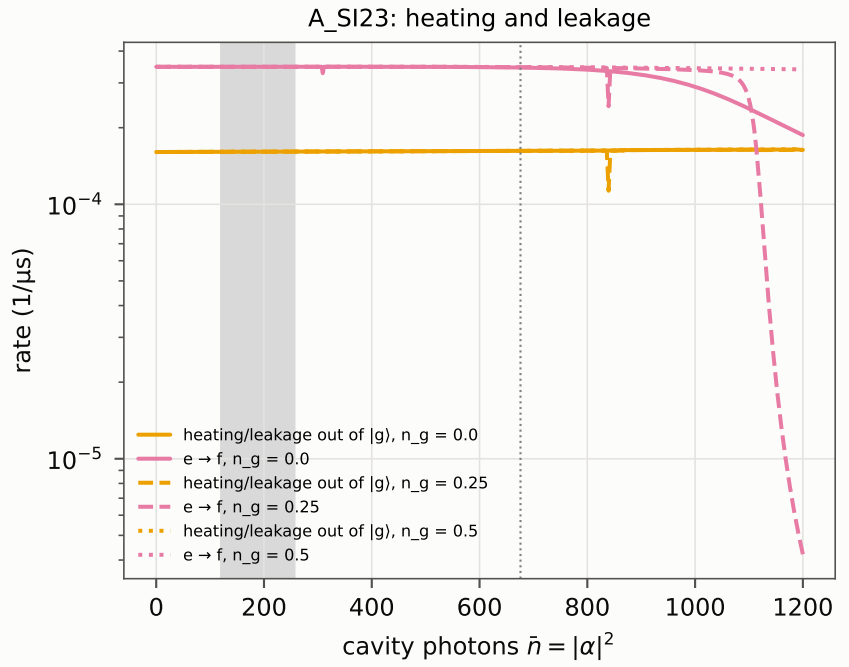
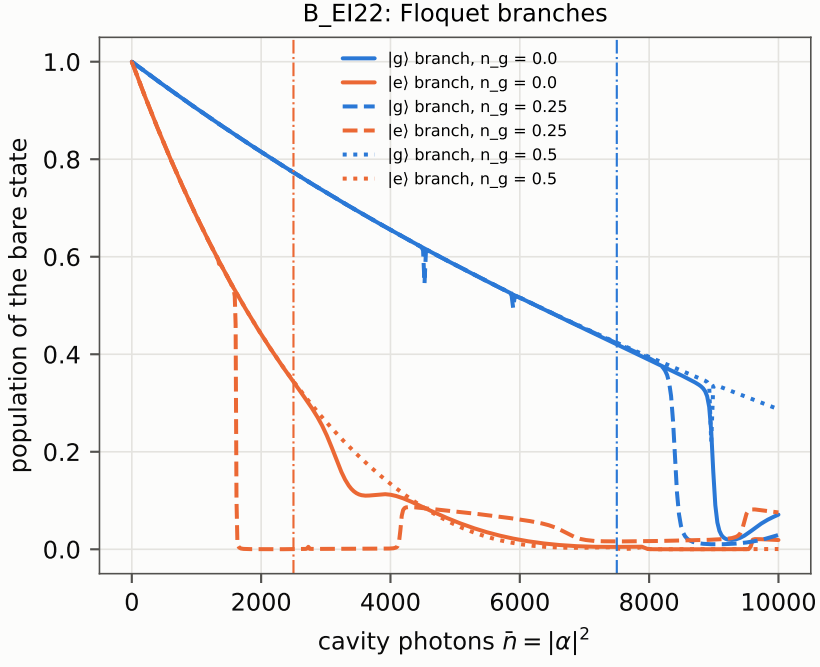
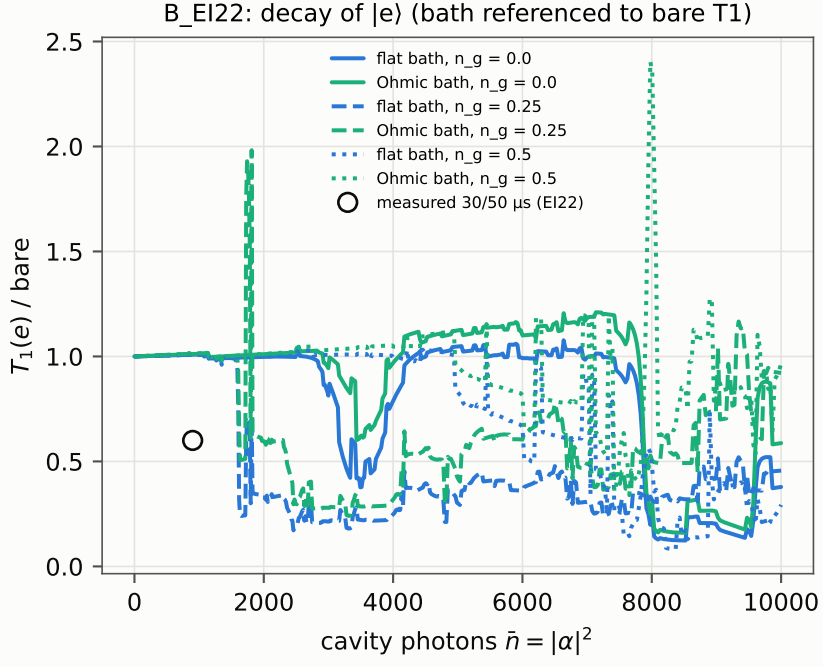
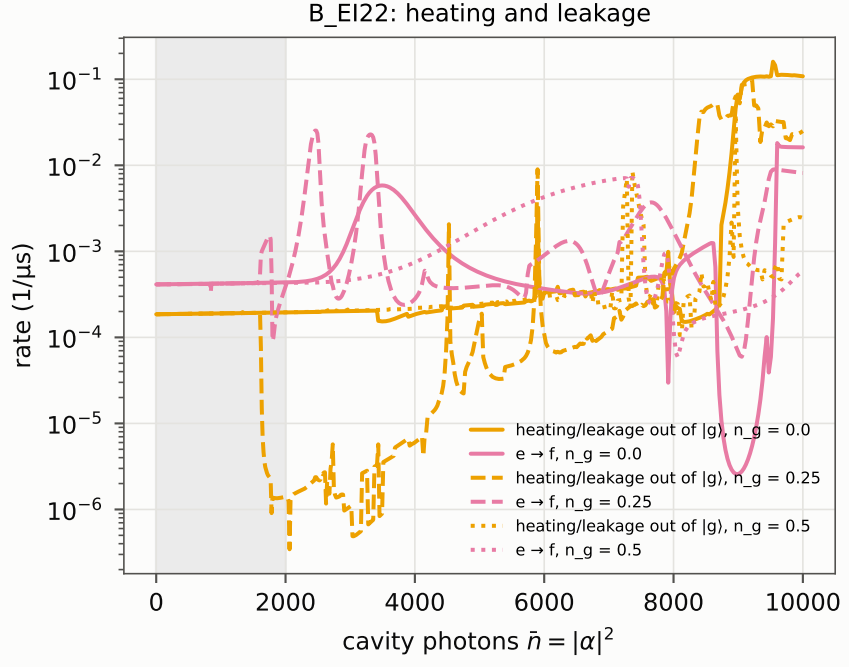

A_SI23: Floquet branches
population of the bare state
cavity photons $\bar{n} = |\alpha|^2$
|g⟩ branch, n_g = 0.0
|e⟩ branch, n_g = 0.0
|g⟩ branch, n_g = 0.25
|e⟩ branch, n_g = 0.25
|g⟩ branch, n_g = 0.5
|e⟩ branch, n_g = 0.5


A_SI23: decay of |e⟩ (bath referenced to bare T1)
$T_1(e)$ / bare
cavity photons $\bar{n} = |\alpha|^2$
flat bath, n_g = 0.0
Ohmic bath, n_g = 0.0
flat bath, n_g = 0.25
Ohmic bath, n_g = 0.25
flat bath, n_g = 0.5
Ohmic bath, n_g = 0.5


A_SI23: heating and leakage
rate (1/μs)
cavity photons $\bar{n} = |\alpha|^2$
heating/leakage out of |g⟩, n_g = 0.0
e → f, n_g = 0.0
heating/leakage out of |g⟩, n_g = 0.25
e → f, n_g = 0.25
heating/leakage out of |g⟩, n_g = 0.5
e → f, n_g = 0.5


B_EI22: Floquet branches
population of the bare state
cavity photons $\bar{n} = |\alpha|^2$
|g⟩ branch, n_g = 0.0
|e⟩ branch, n_g = 0.0
|g⟩ branch, n_g = 0.25
|e⟩ branch, n_g = 0.25
|g⟩ branch, n_g = 0.5
|e⟩ branch, n_g = 0.5


B_EI22: decay of |e⟩ (bath referenced to bare T1)
$T_1(e)$ / bare
cavity photons $\bar{n} = |\alpha|^2$
flat bath, n_g = 0.0
Ohmic bath, n_g = 0.0
flat bath, n_g = 0.25
Ohmic bath, n_g = 0.25
flat bath, n_g = 0.5
Ohmic bath, n_g = 0.5
measured 30/50 μs (EI22)


B_EI22: heating and leakage
rate (1/μs)
cavity photons $\bar{n} = |\alpha|^2$
heating/leakage out of |g⟩, n_g = 0.0
e → f, n_g = 0.0
heating/leakage out of |g⟩, n_g = 0.25
e → f, n_g = 0.25
heating/leakage out of |g⟩, n_g = 0.5
e → f, n_g = 0.5